18

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. in Applied Sciences

B.Sc. Second year Semester II Examination – November/December 2023
CHE 2103 – ANALYTICAL CHEMISTRY

Time: One (01) hour.

Answer both questions.
The use of a non-programmable calculator is permitted.

---

1. a) Solve the following equations using the correct number of significant figures.

   i. 34.683 + 58.930 + 68.35112

   ii, 8.578 / 4.33821

   b) Calculate the absolute uncertainty of the following operations.

$$\frac{(13.67 \pm 0.02)(120.4 \pm 0.2)}{(4.623 \pm 0.006)} = 356.0\ (\pm ?)$$

   c) The percentage tin content of brass sample was analyzed and the following results were obtained. Calculate the standard deviation and the coefficient of variation for this analysis.

   33.27, 33.37, and 33.47

   d) Discuss briefly the following types of error:
   (i). The determinate error.
   (ii). An indeterminate error.

   e) The following replicate molarities were obtained when standardizing a solution: 0.1067, 0.1071, 0.1066 and 1050. Can one of the results be discarded as due to accidental error at 95% confidence level?

No. of observation	Confidence level		
	Q_{90}	Q_{95}	Q_{99}
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821

f) Universal College Board administered an IQ level test to 50 randomly selected academics. The result they found was that the average IQ level score was 120 with a variance of 121. Assume that the t score is 2.407. What is the population mean for this test.

g) Define Systematic sampling and Stratified sampling.

h) Discuss the following used in paper chromatography:

(i). R_f value and the mobile and stationary phases in paper chromatography.

(ii). Factors that affect the R_f value of a compound.

($8 \times 15 = 120$ marks)

2. Answer Part A or Part B

<u>Part A</u>

a) (i). Briefly explain the calibration curve and standard addition methods used in chemical analysis. Give one advantage in each method.

(ii). Write a formula to express the concentration of manganese ([Mn]) in a solution made by mixing 5.00 mL of a sample containing manganese that has a concentration of Cx and 3.00 mL of a standard manganese solution with a concentration of 15.00 ppm diluted to a total volume of 25.00 mL.

(30 marks)

b) Find the line using linear least-squares method and determine the value of x, if y = 4.5.

x	**1**	**2**	**3**	**4**	**5**
y	**2**	**5**	**3**	**8**	**7**

Hint: $m = (n\sum xy - \sum y\sum x)/n\sum x^2 - (\sum x)^2$

$b = (\sum y - m\sum x)/n$

(30 marks)

c) A 0.6128 g sample which contains NaCl and KCl was treated with $AgNO_3$ yielded 1.039 f of dried AgCl (143.2 g mol^{-1}). Determine the percentage of Na and K in the sample.
(The molecular weights of NaCl and KCl are 58.44 g mol^{-1} and 74.55 g mol^{-1} respectively).

(20 marks)

Part B

a) (i). Derive the Henderson–Hasselbalch equation for a buffer solution.

(ii). A buffer solution with a pH of 3.87 was prepared using ethanoic acid and sodium ethanoate. In the buffer solution, the concentration of ethanoate ions was 0.136 mol dm^{-3} Calculate the concentration of ethanoic acid in the buffer solution.

(The value of Ka for ethanoic acid is 1.74×10^{-5} mol dm^{-3} at 25°C).

(20 marks)

b) Calculate the pH values for the titration of 50.00 mL of 0.0500 M HCl at 0.00, 10.00, 25.00. 25.01, and 30 mL addition of 0.100 M NaOH. Construct the titration curve for your results.

(25 marks)

c) (i). Write charge balance expression for the system formed when a 0.010 M NH_3 solution is saturated with AgBr.

(ii). Calculate the molar solubility of $Mg(OH)_2$ in water by systematic approach.

Ksp of $Mg(OH)_2 = 2 \times 10^{-12}$ $(mol\ dm^{-3})^3$

$K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$ $(mol\ dm^{-3})^2$

(35 marks)

--- END ---